

population are the followings: the chi-square distribution, the student's t-distribution, the F-distribution, and the beta distribution. In this chapter, we only consider the first three distributions, since the last distribution was considered earlier.

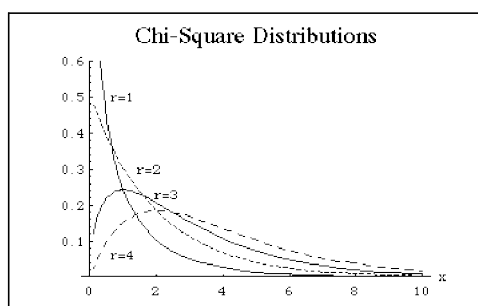
14.1. Chi-square distribution

In this section, we treat the Chi-square distribution, which is one of the very useful sampling distributions.

Definition 14.1. A continuous random variable X is said to have a chi-square distribution with r degrees of freedom if its probability density function is of the form

$$f(x; r) = \begin{cases} \frac{1}{\Gamma(\frac{r}{2}) 2^{\frac{r}{2}}} x^{\frac{r}{2}-1} e^{-\frac{x}{2}} & \text{if } 0 \leq x < \infty \\ 0 & \text{otherwise,} \end{cases}$$

where $r > 0$. If X has chi-square distribution, then we denote it by writing $X \sim \chi^2(r)$. Recall that a gamma distribution reduces to chi-square distribution if $\alpha = \frac{r}{2}$ and $\theta = 2$. The mean and variance of X are r and $2r$, respectively.



Thus, chi-square distribution is also a special case of gamma distribution. Further, if $r \rightarrow \infty$, then chi-square distribution tends to normal distribution.

Example 14.1. If $X \sim GAM(1, 1)$, then what is the probability density function of the random variable $2X$?

Answer: We will use the moment generating method to find the distribution of $2X$. The moment generating function of a gamma random variable is given by

$$M(t) = (1 - \theta t)^{-\alpha}, \quad \text{if } t < \frac{1}{\theta}.$$

Since $X \sim GAM(1, 1)$, the moment generating function of X is given by

$$M_X(t) = \frac{1}{1-t}, \quad t < 1.$$

Hence, the moment generating function of $2X$ is

$$\begin{aligned} M_{2X}(t) &= M_X(2t) \\ &= \frac{1}{1-2t} \\ &= \frac{1}{(1-2t)^{\frac{2}{2}}} \\ &= \text{MGF of } \chi^2(2). \end{aligned}$$

Hence, if X is $GAM(1, 1)$ or is an exponential with parameter 1, then $2X$ is chi-square with 2 degrees of freedom.

Example 14.2. If $X \sim \chi^2(5)$, then what is the probability that X is between 1.145 and 12.83?

Answer: The probability of X between 1.145 and 12.83 can be calculated from the following:

$$\begin{aligned} P(1.145 \leq X \leq 12.83) &= P(X \leq 12.83) - P(X \leq 1.145) \\ &= \int_0^{12.83} f(x) dx - \int_0^{1.145} f(x) dx \\ &= \int_0^{12.83} \frac{1}{\Gamma\left(\frac{5}{2}\right) 2^{\frac{5}{2}}} x^{\frac{5}{2}-1} e^{-\frac{x}{2}} dx - \int_0^{1.145} \frac{1}{\Gamma\left(\frac{5}{2}\right) 2^{\frac{5}{2}}} x^{\frac{5}{2}-1} e^{-\frac{x}{2}} dx \\ &= 0.975 - 0.050 \quad (\text{from } \chi^2 \text{ table}) \\ &= 0.925. \end{aligned}$$

The above integrals are hard to evaluate and thus their values are taken from the chi-square table.

Example 14.3. If $X \sim \chi^2(7)$, then what are values of the constants a and b such that $P(a < X < b) = 0.95$?

Answer: Since

$$0.95 = P(a < X < b) = P(X < b) - P(X < a),$$

we get

$$P(X < b) = 0.95 + P(X < a).$$

We choose $a = 1.690$, so that

$$P(X < 1.690) = 0.025.$$

From this, we get

$$P(X < b) = 0.95 + 0.025 = 0.975$$

Thus, from chi-square table, we get $b = 16.01$.

The following theorems were studied earlier in Chapters 6 and 13 and they are very useful in finding the sampling distributions of many statistics. We state these theorems here for the convenience of the reader.

Theorem 14.1. If $X \sim N(\mu, \sigma^2)$, then $\left(\frac{X-\mu}{\sigma}\right)^2 \sim \chi^2(1)$.

Theorem 14.2. If $X \sim N(\mu, \sigma^2)$ and $X_1, X_2, \dots, X_n$ is a random sample from the population X , then

$$\sum_{i=1}^n \left(\frac{X_i - \mu}{\sigma}\right)^2 \sim \chi^2(n).$$

Theorem 14.3. If $X \sim N(\mu, \sigma^2)$ and $X_1, X_2, \dots, X_n$ is a random sample from the population X , then

$$\frac{(n-1)S^2}{\sigma^2} \sim \chi^2(n-1).$$

Theorem 14.4. If $X \sim GAM(\theta, \alpha)$, then

$$\frac{2}{\theta} X \sim \chi^2(2\alpha).$$

Example 14.4. A new component is placed in service and n spares are available. If the times to failure in days are independent exponential variables, that is $X_i \sim EXP(100)$, how many spares would be needed to be 95% sure of successful operation for at least two years ?

Answer: Since $X_i \sim EXP(100)$,

$$\sum_{i=1}^n X_i \sim GAM(100, n).$$

Hence, by Theorem 14.4, the random variable

$$Y = \frac{2}{100} \sum_{i=1}^n X_i \sim \chi^2(2n).$$

We have to find the number of spares n such that

$$\begin{aligned} 0.95 &= P\left(\sum_{i=1}^n X_i \geq 2 \text{ years}\right) \\ &= P\left(\sum_{i=1}^n X_i \geq 730 \text{ days}\right) \\ &= P\left(\frac{2}{100} \sum_{i=1}^n X_i \geq \frac{2}{100} 730 \text{ days}\right) \\ &= P\left(\frac{2}{100} \sum_{i=1}^n X_i \geq \frac{730}{50}\right) \\ &= P(\chi^2(2n) \geq 14.6). \\ 2n &= 25 \quad (\text{from } \chi^2 \text{ table}) \end{aligned}$$

Hence $n = 13$ (after rounding up to the next integer). Thus, 13 spares are needed to be 95% sure of successful operation for at least two years.

Example 14.5. If $X \sim N(10, 25)$ and $X_1, X_2, \dots, X_{501}$ is a random sample of size 501 from the population X , then what is the expected value of the sample variance S^2 ?

Answer: We will use the Theorem 14.3, to do this problem. By Theorem 14.3, we see that

$$\frac{(501-1)S^2}{\sigma^2} \sim \chi^2(500).$$

Hence, the expected value of S^2 is given by

$$\begin{aligned} E[S^2] &= E\left[\left(\frac{25}{500}\right) \left(\frac{500}{25}\right) S^2\right] \\ &= \left(\frac{25}{500}\right) E\left[\left(\frac{500}{25}\right) S^2\right] \\ &= \left(\frac{1}{20}\right) E[\chi^2(500)] \\ &= \left(\frac{1}{20}\right) 500 \\ &= 25. \end{aligned}$$

Critical values of chi-square (right tail)

Degrees of freedom (df)	Significance level (α)							
	.99	.975	.95	.9	.1	.05	.025	.01
1	-----	0.001	0.004	0.016	2.706	3.841	5.024	6.635
2	0.020	0.051	0.103	0.211	4.605	5.991	7.378	9.210
3	0.115	0.216	0.352	0.584	6.251	7.815	9.348	11.345
4	0.297	0.484	0.711	1.064	7.779	9.488	11.143	13.277
5	0.554	0.831	1.145	1.610	9.236	11.070	12.833	15.086
6	0.872	1.237	1.635	2.204	10.645	12.592	14.449	16.812
7	1.239	1.690	2.167	2.833	12.017	14.067	16.013	18.475
8	1.646	2.180	2.733	3.490	13.362	15.507	17.535	20.090
9	2.088	2.700	3.325	4.168	14.684	16.919	19.023	21.666
10	2.558	3.247	3.940	4.865	15.987	18.307	20.483	23.209
11	3.053	3.816	4.575	5.578	17.275	19.675	21.920	24.725
12	3.571	4.404	5.226	6.304	18.549	21.026	23.337	26.217
13	4.107	5.009	5.892	7.042	19.812	22.362	24.736	27.688
14	4.660	5.629	6.571	7.790	21.064	23.685	26.119	29.141
15	5.229	6.262	7.261	8.547	22.307	24.996	27.488	30.578
16	5.812	6.908	7.962	9.312	23.542	26.296	28.845	32.000
17	6.408	7.564	8.672	10.085	24.769	27.587	30.191	33.409
18	7.015	8.231	9.390	10.865	25.989	28.869	31.526	34.805
19	7.633	8.907	10.117	11.651	27.204	30.144	32.852	36.191
20	8.260	9.591	10.851	12.443	28.412	31.410	34.170	37.566
21	8.897	10.283	11.591	13.240	29.615	32.671	35.479	38.932
22	9.542	10.982	12.338	14.041	30.813	33.924	36.781	40.289
23	10.196	11.689	13.091	14.848	32.007	35.172	38.076	41.638
24	10.856	12.401	13.848	15.659	33.196	36.415	39.364	42.980
25	11.524	13.120	14.611	16.473	34.382	37.652	40.646	44.314
26	12.198	13.844	15.379	17.292	35.563	38.885	41.923	45.642
27	12.879	14.573	16.151	18.114	36.741	40.113	43.195	46.963
28	13.565	15.308	16.928	18.939	37.916	41.337	44.461	48.278
29	14.256	16.047	17.708	19.768	39.087	42.557	45.722	49.588
30	14.953	16.791	18.493	20.599	40.256	43.773	46.979	50.892
40	22.164	24.433	26.509	29.051	51.805	55.758	59.342	63.691
50	29.707	32.357	34.764	37.689	63.167	67.505	71.420	76.154
60	37.485	40.482	43.188	46.459	74.397	79.082	83.298	88.379
70	45.442	48.758	51.739	55.329	85.527	90.531	95.023	100.425
80	53.540	57.153	60.391	64.278	96.578	101.879	106.629	112.329
100	61.754	65.647	69.126	73.291	107.565	113.145	118.136	124.116
1000	70.065	74.222	77.929	82.358	118.498	124.342	129.561	135.807